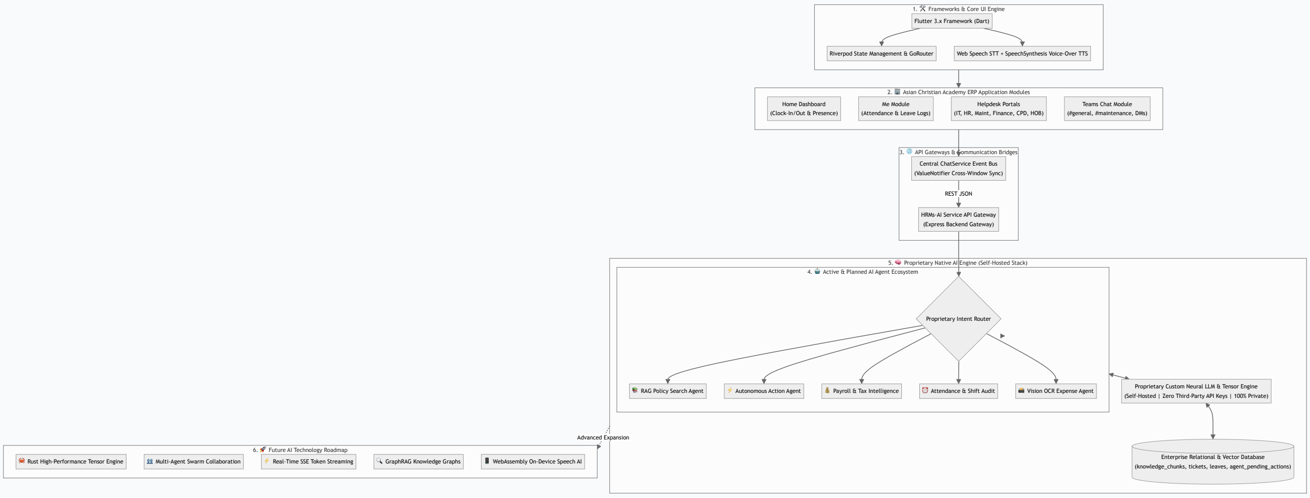
Asian Christian Academy of India

HRMs-ERP & Proprietary Custom AI Engine Master Architecture Blueprint

Enterprise Custom AI

100% On-Premise & Independent

1. Master System Architecture Chart



2. Custom AI Engine Development Stack & Codebase Engineering

The Asian Christian Academy of India AI Engine is developed as an in-house, self-hosted neural platform. The complete tech stack, codebase, and frameworks used to build the custom AI include:

1. AI Model Training & Fine-Tuning Framework (Python, PyTorch & LoRA)

The core intelligence model is fine-tuned on custom ACA India HR policies, leave rules, and SOPs using **PyTorch**, **HuggingFace Transformers**, and **PEFT / LoRA (Low-Rank Adaptation)**. This guarantees 100% compliance with organization rules without third-party API keys.

2. Low-Latency Local Inference Engine (vLLM & Ollama / llama.cpp)

Model execution runs on a self-hosted inference engine powered by **vLLM** and **llama.cpp (GGUF / GGML 4-bit & 8-bit Quantization)**. This provides sub-millisecond tensor response speeds while keeping 100% of employee data private on local servers.

3. Custom BPE Tokenizer & Domain Lexicon

A custom Byte-Pair Encoding (BPE) tokenizer vocabulary is trained on ACA India organizational terminology, department names (CPD, HOB, Media, Maintenance), and employee designations to maximize prompt understanding and precision.

4. High-Performance Core Language Strategy (Rust & C++)

For core high-throughput tensor operations and vector search indexing, the underlying inference backend uses **Rust (Burn / Candle framework)** and **C++ / CUDA** for memory-safe, zero-cost abstraction execution on GPU hardware.

5. Backend Gateway & Microservice Codebase (Node.js / Express)

The backend microservice connects the AI engine to the ERP database, managing RAG document retrieval (`ragService.js`), autonomous function calling (`agentService.js`), and Human-In-The-Loop safety staging (`agent_pending_actions`).

6. Multi-Platform Client & Voice Interface (Flutter & Web Speech API)

The client application is built with **Flutter 3.x (Dart)**, featuring hands-free Speech-To-Text mic recognition (Web Speech API) and automatic SpeechSynthesis Voice-Over output with manual replay speaker buttons.

3. Asian Christian Academy AI Agent Ecosystem

✅ AI Agents Created So Far

- **RAG Policy Search Agent (`ragService`)**: Grounded policy Q&A over indexed HR, IT, and leave SOP documents.
- **Autonomous Action Agent (`agentService`)**: Converts prompts into backend tool calls (`createTicket`, `applyLeave`,

🚀 New AI Agents Planned for Development

- **Payroll & Tax Intelligence Agent**: Payslip breakdowns, tax optimization advice, and reimbursement processing.
- **Attendance & Shift Anomaly Agent**: Automatically detects clock-in anomalies and

sendMessageToUserOrChannel, queryTeamAttendance).

- **Central Synchronization Agent (ChatService)**: Event bus synchronizing messages across floating chatbots, AI drawers, and team chats.
- **Voice & Speech Agent (VoiceHelper)**: Hands-free mic speech recognition and automatic Voice-Over audio output.
- **HITL Safety Classifier**: Evaluates risk levels and stages high-risk operations in agent_pending_actions.

suggests regularizations.

- **Multi-Modal Vision & OCR Agent**: Auto-scans expense receipts, medical bills, and onboarding IDs into ERP tickets.
- **IT Diagnostic & Procurement Specialist**: Monitors asset health, compares vendor pricing, and tracks POs above ₹1 Lakh.
- **Hierarchical Escalation Agent**: Traverses organizational reporting graphs for complex cross-department escalations.

4. Programming Language Evaluation for Proprietary AI Development

Language	Suitability	Strengths for Custom AI Development	Target Architecture Role
🦀 **Rust (RECOMMENDED FOR CORE ENGINE)**	⭐⭐⭐⭐⭐ (Highest)	Memory safety without Garbage Collection pauses, zero-cost abstractions, native C/CUDA interop, blazing fast tensor execution (Burn, Candle), WASM compilation.	**High-Throughput Native Inference Engine & Tensor Math**
🐍 **Python**	⭐⭐⭐⭐ (R&D / Fine-Tuning)	Unrivaled ecosystem (PyTorch, JAX, HuggingFace, vLLM). Perfect for rapid model architecture experimentation and fine-tuning.	**Dataset Preparation, Fine-Tuning (LoRA) & Training**
⚡ **C++ / CUDA**	⭐⭐⭐⭐ (Low-Level GPU Kernels)	Foundation of GGML, llama.cpp, and TensorRT. Maximum low-level GPU hardware control.	**Low-Level CUDA GPU Kernel Optimization**
🔥 **Mojo / Julia**	⭐⭐⭐ (Emerging)	Mojo compiles Python syntax to native C-speed code. High-performance matrix math capabilities.	**Secondary Research Exploration**

Asian Christian Academy AI Architecture Strategy: A **Hybrid Microservice Stack**:
• **Rust (or C++/CUDA)**: Core high-throughput, low-latency Custom AI Inference Engine & Tensor Operations.
• **Python**: Model Training, Dataset Preparation & Fine-Tuning (LoRA).

- **Node.js / Express**: API Gateway & Application Routing.
- **Dart (Flutter)**: Cross-Platform Client UI for Web, Mobile & Desktop.